FROM TOKENS TO ENERGY: THE COST OF REPLACING HUMAN ENGINEERING WITH AI

Ge Zhang
Department of Energy Science & Engineering
Stanford University
gmzhang@stanford.edu

Anthony R. Kovscek
Department of Energy Science & Engineering
Stanford University
kovscek@stanford.edu

ABSTRACT

Artificial intelligence (AI) systems are demonstrating increasing capability at software engineering tasks, raising the prospect of large-scale substitution of human engineering labor by AI inference. In this paper, we construct a bottom-up quantitative model linking engineering labor economics to AI compute infrastructure requirements, energy consumption, and capital market opportunity across a spectrum of AI human resource substitution scenarios. Firstly, the *token* is taken as the analytical bridge between labor economics (FTE-hours, compensation) and computational economics (FLOP/s, kilowatt-hours). Using a six-category task taxonomy to bound substitutability, empirical token consumption data from agentic coding benchmarks [Bai et al., 2026], and hardware throughput benchmarks for current AI tools, we derive the compute and energy requirements for replacing 10%, 30%, and 80% of the global software engineering workforce (assuming 24 million FTEs) under fully agentic AI deployment. To keep an serious estimation about 30% substitution, the fully agentic scenario use approximately 28.8 trillion tokens per day, requiring roughly one hundred thousand H100-class GPUs and 110 MW of data center facility power, which is equivalent to a single mid-size power facility. Construction capex is approximately $1.65 billion; annual energy cost approximately $58 million; fully-loaded AI cost approximately $9,020 per replaced FTE-year, or 6.8% of a US software developer's median salary of $133,080 [Bureau of Labor Statistics, 2025]. Annual labor savings at 30% substitution are approximately $511 billion. The engineering substitution demand represents 0.10% of the IEA's projected 2030 global data center load of 107.9 GW, which is unexpectedly low. However, the binding constraints on AI engineering deployment are not economic but operational: current agentic capability (SWE-bench resolution rate), institutional adoption friction, and grid interconnection timelines of 2–5 years for large loads. We map the resulting opportunity and tasks across energy companies, data center developers, equipment suppliers, and capital markets instruments, and identify serval priorities on fast-track grid interconnection for AI-critical loads, pre-funded engineering workforce transition programs, and mandatory AI inference carbon accounting.

Keywords: artificial intelligence, engineering automation, token consumption, data center energy, AI infrastructure, labor substitution, energy markets

Contents

1 Introduction and Motivation **3**
1.1 From Engineering Labor to Tokenized Cognition 3

2 Related Work **4**

3 Methodology **5**
3.1 Overview 5

3.2 Model Architecture . . . 5
3.3 Key Data Sources . . . 6
3.4 Assumptions and Limitations . . . 6
3.4.1 Locked Assumptions . . . 6
3.4.2 Pre-Publication Verification Requirements . . . 7
3.4.3 Model Limitations . . . 7

4 Results **8**
4.1 Engineering Labor Substitution and Token Demand . . . 8
4.1.1 A Taxonomy of Engineering Work by Token-Substitutability . . . 8
4.1.2 The Token Generation Model . . . 9
4.1.3 Aggregate Token Demand as a Function of Substitution Rate . . . 10
4.1.4 Sensitivity Analysis . . . 11
4.2 Compute Infrastructure Requirements . . . 11
4.2.1 From Tokens to FLOPs: The Arithmetic of Inference . . . 11
4.2.2 Hardware Throughput and GPU Requirements . . . 11
4.2.3 Data Center Physical Requirements . . . 12
4.2.4 Phased Deployment Trajectory . . . 12
4.3 Energy Consumption and Economic Cost . . . 13
4.3.1 Total Power Budget . . . 13
4.3.2 Energy Mix and Carbon Intensity . . . 13
4.3.3 Fully-Loaded Cost Per Engineering FTE-Year . . . 13

5 Discussion **13**
5.1 Energy Market Implications . . . 14
5.1.1 Grid Load Growth and Infrastructure Demand . . . 14
5.1.2 Power Source Requirements and Decarbonization Tension . . . 14
5.2 Economic Comparison: AI vs. Human Engineering Labor . . . 15
5.2.1 Cost Structure and Geographic Breakeven . . . 15
5.2.2 Labor Savings and Substitution Economics . . . 15
5.2.3 Non-Economic Constraints on Adoption . . . 16
5.3 Environmental and Workforce Considerations . . . 16
5.3.1 Environmental Footprint and Water Constraints . . . 16
5.3.2 Workforce Displacement at Scale . . . 16

6 Conclusion **17**
6.1 Summary of Findings . . . 17
6.2 Policy Recommendations . . . 17
6.3 Future Research Directions . . . 18

A Full Parameter Table **21**

B Sensitivity Analysis **21**

C Glossary **22**

D Timeline of AI Engineering Tools (2020–2025) **22**

1 Introduction and Motivation

1.1 From Engineering Labor to Tokenized Cognition

Every major wave of engineering automation has historically mechanized physical execution while preserving human cognition as the central coordinating function. Computer-aided design (CAD) systems beginning in the 1970s eliminated manual drafting as a distinct profession within architecture and civil engineering, yet total employment in those industries expanded as lower design costs increased the scale of construction and infrastructure development. Electronic design automation (EDA) tools in the 1980s and 1990s removed the need for manual integrated circuit layout, while simultaneously enabling exponential growth in semiconductor complexity and production volume. In each case, automation primarily targeted procedural or physically mediated labor, while abstraction, system design, trade-off analysis, and novel problem formulation remained fundamentally human activities.

Large language model (LLM) systems represent a structural break from this historical pattern. Unlike prior automation technologies, contemporary AI systems increasingly target linguistic and cognitive components of engineering labor itself. Recent advances in autonomous and agentic software engineering systems allow language models not only to generate code fragments, but also to plan execution strategies, invoke tools, navigate repositories, interpret compiler feedback, execute debugging loops, and iteratively refine software artifacts with limited direct human intervention [Wang et al., 2025, Patil et al., 2025].

The economic significance of this transition is substantial. Software engineering represents one of the most economically valuable categories of skilled labor in the global economy. The United States alone employed approximately **1.89 million software developers** in 2024 at a median annual wage of roughly **$131,450**, with employment continuing to grow rapidly according to Bureau of Labor Statistics [2025]. Globally, the professional software developer population reached approximately **20.8 million** in 2025 according to JetBrains [2025], excluding many hobbyist and student programmers, while broader estimates place the worldwide developer population between **27–29 million** [Evans Data Corporation, 2025]. Assuming global software engineering compensation averages approximately 50% below U.S. median wage levels because of international labor market differentials, these estimates imply an annual compensation pool of approximately **$1.4 trillion** for professional software developers alone, and approximately **$1.8–1.9 trillion** under broader definitions of the global developer workforce.

The current state of AI capability establishes a meaningful lower bound on the technical feasibility of software engineering automation. As of early 2026, frontier coding agents such as *Claude Opus 4.5* achieved approximately **77–81% resolution rates on SWE-bench Verified**, a widely used benchmark evaluating repository-level software maintenance and bug-fixing tasks under executable testing conditions [SWE-bench, 2026, Anthropic, 2025]. Although benchmark performance does not directly imply full occupational replacement, these results demonstrate that language models can already execute substantial fractions of practical software maintenance workflows without requiring direct human-authored code generation.

At the same time, recent research increasingly frames AI agents as economically meaningful actors rather than isolated software tools [Rothschild et al., 2025]. This shift creates a fundamental analytical challenge. Labor economics measures productive activity in units such as **FTE-hours**, **headcount**, and **compensation dollars**. AI infrastructure economics instead measures productive activity in **tokens**, **FLOPs**, **GPU-hours**, and **electrical power consumption**. Existing literature generally analyzes either labor substitution or compute infrastructure scaling in isolation, leaving the quantitative relationship between cognitive labor replacement and physical compute demand poorly understood.

This paper argues that the missing bridge between these domains is the **token**: the atomic unit of language-model computation. Tokens provide a measurable intermediate representation linking engineering work to computational and energy requirements. Token counts are:

- directly measurable through API billing and inference logs,
- directly connected to computation through transformer scaling relationships,
- directly connected to economic cost through published API pricing,
- and directly connected to physical infrastructure requirements through GPU utilization and electrical power demand.

Recent work on "tokenomics" in agentic software engineering demonstrates that autonomous coding systems can consume orders of magnitude more tokens than conventional copilot-style interaction patterns [Salim et al., 2026, Bai et al., 2026]. Contemporary copilot-assisted programming workflows appear to operate on the order of $\mathbf{10^5}$ tokens per engineer per day, whereas fully agentic coding systems evaluated on SWE-bench-style tasks may consume on the order of $\mathbf{10^6}$–$\mathbf{10^7}$ tokens per complex software task. This distinction is economically and physically important: modest increases in coding productivity require relatively small increases in inference demand, while large-scale replacement

of engineering labor with autonomous AI systems may imply a qualitative transformation in global compute and energy consumption.

The AI substitution hypothesis examined in this paper is therefore not binary. We do not argue that AI will categorically replace all software engineers. Instead, we model a *substitution rate*: the fraction of engineering full-time-equivalent (FTE) labor hours effectively executed by AI inference systems at production-acceptable quality. McKinsey Global Institute estimates that generative AI could automate or significantly augment activities representing approximately **60–70% of employee work time** across the global economy under full deployment scenarios [McKinsey Global Institute, 2023]. For software engineering specifically, a profession highly concentrated in text-mediated and cognitive workflows, the technical ceiling may be substantially higher than economy-wide averages. Our task-weighted taxonomy analysis suggests a theoretical upper bound of approximately **71%** substitutable engineering FTE-hours under advanced agentic deployment assumptions.

The central scenario modeled throughout this paper assumes approximately **30% effective engineering labor substitution**, broadly anchored to accelerated AI deployment scenarios in STEM-related occupations projected by McKinsey Global Institute [McKinsey Global Institute, 2024]. Under this scenario, software engineering labor increasingly becomes a form of tokenized cognition executed through GPU-based inference systems. The central question examined in this paper is therefore not merely whether AI can replace portions of software engineering labor, but rather:

> *How much computational infrastructure, electrical power, and industrial capacity are required to replace human cognitive engineering labor at global scale?*

The following part of this paper proceeds as, Section 2 reviews prior literature on agentic software engineering, AI labor substitution, token economics, and AI infrastructure scaling. Section 3 presents the quantitative framework linking engineering labor substitution to token demand, compute requirements, and energy consumption. Section 4 evaluates token throughput, GPU infrastructure requirements, power consumption, and cost under multiple AI deployment scenarios, including the central assumed 30% engineering labor substitution case. Section 5 examines broader economic, infrastructure, environmental, and policy implications arising from large-scale AI-mediated software engineering. Finally, Section 6 summarizes the paper's analysis and outlines directions for future research.

2 Related Work

This paper sits at the intersection of four active research areas: AI-driven labor automation, foundation-model scaling and inference economics, agentic software engineering systems, and AI infrastructure energy demand. Existing work has extensively examined each layer independently, but relatively little research quantitatively connects labor substitution to compute infrastructure and energy consumption through tokenized inference demand.

Research on automation and labor economics has long analyzed how technological systems substitute for human tasks rather than entire occupations [Autor et al., 2003, Acemoglu and Restrepo, 2022]. Recent work on generative AI extends this framework to knowledge-intensive and cognitive labor. McKinsey Global Institute projects that generative AI could automate or significantly augment activities representing 60–70% of employee work time under full deployment scenarios [McKinsey Global Institute, 2023], while Goldman Sachs estimates substantial labor-market exposure across advanced economies [Goldman Sachs Global Investment Research, 2023]. However, these analyses primarily focus on labor-market and productivity outcomes without modeling the computational infrastructure required to operationalize AI substitution at scale.

A parallel literature examines the scaling behavior and computational economics of large language models. Foundational work on transformer scaling laws demonstrated predictable relationships between model size, data, and compute requirements [Kaplan et al., 2020, Hoffmann et al., 2022]. Subsequent foundation-model research established the broad capability generalization of large language models across linguistic and reasoning tasks [Brown et al., 2020, Bommasani et al., 2021]. These works characterize the computational properties of frontier models, but generally abstract away downstream labor and industrial implications.

More recent work studies autonomous and agentic AI systems capable of multi-step reasoning, tool invocation, and software engineering workflows [Wang et al., 2025, Patil et al., 2025]. SWE-bench and related benchmarks operationalize repository-level software maintenance evaluation for coding agents [Jimenez et al., 2024]. Recent empirical studies further quantify the token economics of autonomous coding systems, showing that agentic software engineering workflows may consume orders of magnitude more tokens than conventional interactive prompting [Salim et al., 2026, Bai et al., 2026]. This emerging literature establishes token consumption as an important operational and economic property of AI agents, but has not yet connected token demand to large-scale labor substitution scenarios.

Finally, a growing infrastructure and energy literature examines the physical footprint of modern AI systems. The International Energy Agency projects rapid growth in global data-center electricity demand driven by AI deployment

[International Energy Agency, 2024], while industry and systems analyses study GPU efficiency, inference throughput, and datacenter scaling constraints [Lawrence Berkeley National Laboratory, 2024, NVIDIA Corporation, 2024c]. At the same time, recent economic work increasingly frames AI agents as autonomous economic actors operating within broader digital production systems [Rothschild et al., 2025]. Yet existing energy and infrastructure analyses rarely specify which categories of cognitive labor generate the inference demand underlying these projections.

This paper contributes a quantitative bridge across these literatures. We model software-engineering labor substitution in token terms, linking engineering FTE-hours to token demand, GPU infrastructure requirements, electrical power consumption, and deployment cost through a unified bottom-up framework. In doing so, the paper reframes AI labor substitution not only as an economic phenomenon, but also as an infrastructure and energy systems problem.

3 Methodology

3.1 Overview

The paper constructs a five-layer bottom-up quantitative model that connects engineering labor economics to AI compute infrastructure requirements. The model is deterministic given its inputs, making all intermediate calculations fully reproducible from the locked assumptions documented in Section 3.4. We run the model across three substitution rates (10%, 30%, 80%) and two deployment modes (Copilot-augmented and Fully Agentic), yielding six primary scenario combinations. Results are reported in Section 4 (encompassing token demand, compute infrastructure, and energy cost analysis); market and risk implications are discussed in Section 5.

3.2 Model Architecture

This paper constructs a **five-layer bottom-up quantitative model** connecting engineering labor economics to compute infrastructure requirements to energy consumption and cost. Each layer produces outputs that serve as inputs to the next. The model is built to be auditable: every transformation is explicit, every locked assumption is documented, and every empirical anchor is tagged with its evidentiary quality.

Layer 1 — Labor Input Model

Inputs: Global software engineering workforce size — centrally estimated at 24 million FTEs **[SECONDARY]** [JetBrains, 2025, Evans Data Corporation, 2025]; total engineering workforce of approximately 33–34 million FTEs derived using the US BLS architecture-and-engineering-to-software-developer employment ratio of 1.39× **[PRIMARY]** [Bureau of Labor Statistics, 2025]; substitution rate (10% conservative / 30% central / 80% aggressive).

Key transformation: Substitution rate × software engineering workforce = FTE-equivalents whose work hours are effectively replaced by AI inference. The taxonomy of engineering work by token-substitutability (Section 4.1.1, Table 8) yields a task-category-weighted effective replacement ceiling of approximately 71% of FTE-hours for software engineering.

Output: Token-active FTEs by scenario (2.4 million at 10%; 7.2 million at 30%; 19.2 million at 80%).

Primary uncertainty: Workforce size carries a ±20% range from two credible but methodologically distinct published estimates. The substitution rate is the model's dominant uncertainty.

Layer 2 — Token Demand Model

Inputs: Token-active FTEs from Layer 1; deployment mode (Copilot-augmented: 120,000 tokens/day/FTE **[ESTIMATE]**; Agentic: 4,000,000 tokens/day/FTE **[SECONDARY]** from Bai et al. [2026]).

Key transformation: FTEs × tokens/day/FTE = total daily token demand.

Output: Daily token demand by scenario and deployment mode, ranging from 288 billion tokens/day (10% substitution, Copilot mode) to 76.8 trillion tokens/day (80% substitution, Agentic mode). The central scenario (30% substitution, Agentic mode) produces **28.8 trillion tokens/day**.

Primary uncertainty: Per-task token consumption (1 million–3.5 million range; 3.5× spread) combined with tasks-per-day uncertainty (1–4 range) creates a combined 14× uncertainty band in aggregate token demand.

Layer 3 — Compute Model

Inputs: Daily token demand from Layer 2; GPU throughput benchmarks (H100 SXM5: 3,311 tokens/second at 70B model scale, FP16, batch size 64 **[SECONDARY]** [Hyperstack.cloud, 2024]; B200: 60,000 tokens/second **[PRIMARY]** [NVIDIA Corporation, 2024a]); PUE of 1.56 **[SECONDARY]** [Uptime Institute, 2024]; H100 inference TDP of 700 W/GPU **[PRIMARY]** [NVIDIA Corporation, 2024b].

Key transformation:

tokens/day $\div 86{,}400$ sec/day = tokens/sec $\rightarrow \div$GPU throughput = GPU count $\rightarrow \times 700\text{W} \times$PUE = facility power (MW)

Output: Central scenario: approximately 101,000 H100 GPUs; 110 MW total facility power. Equivalent Blackwell-generation figures: approximately 5,550 B200 GPUs; 77 GB200 NVL72 racks.

Layer 4 — Energy and Cost Model

Inputs: Facility power in MW from Layer 3; construction cost of $15 million/MW **[ESTIMATE]**; electricity price of $60/MWh blended **[ESTIMATE]**; amortization period of 20 years; API token cost of $9/MTok blended frontier pricing **[ESTIMATE]**.

Output: Fully-loaded AI cost per FTE-year of approximately **$9,020** in the central scenario, equal to approximately 6.8% of the US software developer median salary of $133,080 **[PRIMARY]** [Bureau of Labor Statistics, 2025].

Layer 5 — Market Sizing Model

Inputs: Facility power; GPU counts; annual energy costs; PPA prices; public REIT and equipment supplier revenues.

Output: Investment thesis sizing and framing for energy companies, data center developers, equipment suppliers, commodity markets, and capital markets participants (Section 5).

Table 1: Five-Layer Model Summary

Layer	Name	Key Inputs	Outputs	Primary Uncertainty
1	Labor Input	Workforce; substitution rate	Token-active FTEs	Rate speculative; ś20% workforce
2	Token Demand	FTEs; tok/FTE/day	288B–76.8T tok/day	14Œ combined (task tok + tasks/day)
3	Compute	Token demand; H100 throughput	33,700–269,000 H100s	ś30% throughput benchmark
4	Energy & Cost	Power; $/MW; $/MWh; API $	$9,020/FTE-yr central	$20–$120/MWh elec. range
5	Market Sizing	All of above + PPA prices	Investment thesis sizing	GS $1.15T capex needs **[SECONDARY]** [Goldman Sachs

The model runs across three substitution rates (10%, 30%, 80%) and two deployment modes (Copilot: 120,000 tokens/day/FTE; Agentic: 4,000,000 tokens/day/FTE), generating six primary scenario combinations. The scenarios are not forecasts of what will happen. They are structured inputs that allow the infrastructure implications of a given degree of AI engineering deployment to be read directly from the tables.

3.3 Key Data Sources

The paper draws on four primary data domains documented in Tables 2—5. Sources marked **[VERIFY]** require retrieval of the original primary document before publication.

Table 2: Domain 1 — Engineering Workforce and Compensation

Source	Claims Supported	Quality
BLS OES May 2024	US SW developer count (1.847M); median wage ($133,080)	**[PRIMARY]**
JetBrains Dev Ecosystem 2025	Global developers: 19.6M	**[SECONDARY]**
Evans Data / Statista	Global developers: 28.7M	**[SECONDARY]**
McKinsey MGI May 2024	30% STEM automation by 2030	**[SECONDARY]**
McKinsey MGI Nov 2025	57% technically automatable	**[SECONDARY]** **[VERIFY]**
GitHub Copilot docs	20M users; 55% speed improvement; 30% acceptance	**[SECONDARY]**

3.4 Assumptions and Limitations

3.4.1 Locked Assumptions

During the research phase, five cases of conflicting or ambiguous empirical data were identified where two or more credible published sources reported materially different values for the same parameter. In each case, a single working value was selected.

Table 3: Domain 2 — Token and Compute Benchmarks

Source	Claims Supported	Quality
Bai et al. 2026, arXiv:2604.22750	1–3.5M tok/task; 30× variance; 85:15 ratio	**[SECONDARY]**
NVIDIA H100 SXM5 product brief	989 TFLOPS FP16; 700W TDP	**[PRIMARY]**
NVIDIA DGX SuperPOD Design Guide	40–48 kW/rack air-cooled	**[PRIMARY]**
NVIDIA Blackwell InferenceMAX	60,000 tok/s per B200	**[PRIMARY]**
Hyperstack.cloud benchmark	3,311 tok/s per H100 SXM5 (Llama 3.1 70B)	**[SECONDARY]**
Epoch AI scaling laws	140 GFLOPs/token (70B model, 2N rule)	**[SECONDARY]**
Anthropic Claude API docs	$3/M input, $15/M output (Sonnet 4.6)	**[PRIMARY]**
OpenAI API pricing	$3/M input, $12/M output (GPT-4.1)	**[PRIMARY]**
SWE-bench leaderboard / Cognition	Devin 2.0: 13.86% resolution rate	**[SECONDARY]** **[VERIFY]**

Table 4: Domain 3 — Data Center Energy and Infrastructure

Source	Claims Supported	Quality
IEA "Energy and AI" (April 2025)	415 TWh (2024); 945 TWh (2030 Base Case); AI server +30%/yr	**[PRIMARY]**
LBNL 2024 Data Center Report	US: 183 TWh (2024); 325–580 TWh by 2028	**[PRIMARY]**
Uptime Institute Survey 2024	Global PUE: 1.56	**[SECONDARY]**
AWS Sustainability Report 2024	WUE: 0.19 L/kWh	**[SECONDARY]**
Cushman & Wakefield 2024	$10–12M/MW standard hyperscale	**[SECONDARY]**
Data Center Dynamics	$20M+/MW AI-optimized	**[SECONDARY]**
Microsoft, Google, Amazon, Meta SEC filings	Combined 2025 capex guidance ∼$448B	**[PRIMARY]**
NVIDIA quarterly earnings	FY2025 DC revenue $116.2B	**[PRIMARY]**

3.4.2 Pre-Publication Verification Requirements

3.4.3 Model Limitations

The following factors are not modeled in this paper and represent the primary sources of real-world deviation from the projections presented in Sections 4–5:

1. **Model efficiency improvements.** The paper uses H100 SXM5 hardware and 2025-vintage frontier model architectures as the baseline throughout. Blackwell-generation hardware already delivers approximately 18× GPU count reduction per workload at the same model scale.

2. **Heterogeneous workload mixing across model sizes.** The paper uses the 70B model class as the uniform baseline. In practice, a significant fraction of engineering tasks can be handled by smaller models (7B–13B parameters) at dramatically lower compute and cost per token.

3. **Geographic variation in energy prices and grid carbon intensity.** The paper uses $60/MWh as a blended electricity price for all scenarios. Actual prices range from approximately $20/MWh in hydro-rich regions to $120+/MWh in constrained markets.

4. **Institutional adoption friction.** The substitution rate scenarios do not account for procurement cycles (typically 6–18 months), security review, legal liability for AI-generated code, or the trust deficit that must be overcome before production engineering workflows are delegated to AI agents.

Table 5: Domain 4 — Energy Markets and Capital Markets

Source	Claims Supported	Quality
Constellation Energy / ANS / Utility Dive	Microsoft–TMI-1 PPA: 835 MW, 20-year, 2027 restart	**[PRIMARY]**
Bloomberg Intelligence / Jefferies	TMI-1 PPA price: ∼$100–112/MWh	**[SECONDARY]**
Talen Energy / ESG Today	Amazon–Susquehanna: $650M, 1,920 MW through 2042	**[PRIMARY]**
About Amazon / ESG Today	Amazon–Energy Northwest SMR: 320–960 MW	**[PRIMARY]**
LevelTen Energy PPA Price Index Q3 2024	Solar: $56.58/MWh; wind: $65.63/MWh	**[SECONDARY]**
TradeTech / ANS	Uranium: $72.63/lb year-end 2024	**[PRIMARY]**
Equinix Q4 2024 earnings	Revenue $8.748B; 365 MW xScale leased	**[PRIMARY]**
Digital Realty Q4 2024 earnings	9.1M sq ft under construction; $100M Q4 bookings	**[PRIMARY]**
Vertiv Q4 2024 earnings	Revenue $8.0B (+17%); organic orders +30%	**[PRIMARY]**
Goldman Sachs Insights 2024	$1.15T cumulative AI capex 2025–2027	**[SECONDARY]**
Grid Strategies 2025	>160 GW committed US large loads; 2–5 yr interconnection	**[SECONDARY]**

Table 6: Locked Assumptions — Conflicts and Decisions

Assumption	Locked Value	Range	Rationale
H100 inference power draw	700 W/GPU	700–1,275 W	700 W reflects inference TDP, not full training load
Data center construction cost	$15M/MW	$10–20M/MW	Midpoint of AI-optimized liquid-cooled range
Global SW developer count	24 million	19.6M–28.7M	Midpoint of JetBrains and Evans Data estimates
Agentic token multiplier	50–500× (enterprise)	50–1,000×	Different baselines; 1,000× vs. single-turn; 50–500× vs. multi-turn session
Data center copper demand	375,000 t/yr	375K–1.1M t/yr	Macquarie (data centers only) vs. Sprott (broader AI supply chain)

5. **Jevons dynamics in software production.** The model assumes AI substitution reduces engineering headcount while holding software output approximately constant. An alternative scenario is that AI augmentation enables a fixed number of human engineers to produce substantially more software at lower marginal cost.

4 Results

4.1 Engineering Labor Substitution and Token Demand

4.1.1 A Taxonomy of Engineering Work by Token-Substitutability

Before constructing a token demand model, we must answer a prior question: which engineering activities are actually substitutable by language-model inference? Not all engineering labor translates into text tokens. The correct unit of substitution is not the engineer, but the *task*, and tasks vary enormously in their amenability to token-based automation.

We propose a six-category taxonomy derived from empirical time-use studies of software engineering work [Meyer et al., 2019, Sadowski et al., 2015] and extended to cover broader engineering disciplines.

Deriving the Theoretical FTE Substitution Ceiling

Using the task-weighted effective replacement rates from Table 8:

Table 7: Pre-Publication Verification Checklist

Claim	Current Source	Verification Action
1–3.5M tok/task; 30× variance; 85:15 ratio	Bai et al. 2026 arXiv:2604.22750 (confirmed; not yet peer-reviewed)	Confirm range holds across models; supplement with github.com/LongjuBai/agent_token_consumption_analysis
57% of work hours technically automatable	McKinsey MGI Nov 2025	Retrieve original MGI report before submission; currently cited via secondary summary
OpenAI ∼10B tokens/day (late 2023)	Unverified	Remove from paper; no primary source found

Table 8: Taxonomy of Engineering Work by Token-Substitutability

Work Category	Hours	Substitutability	Eff. Rate	Representative Tasks
Code generation & review	∼35%	High	∼90%	Feature impl., PR review, debugging
Documentation & specification	∼15%	Very High	∼95%	RFCs, API docs, ADRs, test plans
Testing & validation	∼15%	High	∼85%	Unit/integration test generation
Design & architecture	∼20%	Medium	∼40%	System design, trade-off analysis
Communication & coordination	∼10%	Medium	∼45%	Meeting summaries, ticket triage
Physical/embodied work	∼5%	Low	∼5%	PCB rework, lab measurements

$$\text{FTE ceiling} = \sum_k w_k \times r_k = (0.35\times0.90)+(0.15\times0.95)+(0.15\times0.85)+(0.20\times0.40)+(0.10\times0.45)+(0.05\times0.05) = \mathbf{0.714}$$

The taxonomy-derived ceiling is approximately **71% of engineering FTE-hours**.

4.1.2 The Token Generation Model

Per-FTE Token Demand: Two Canonical Modes

Mode A — Copilot-Augmented (human-in-the-loop): A representative chat-heavy engineering workday involves approximately 10 substantive AI interactions, each consuming around 10,000 input tokens and generating 2,000 output tokens **[ESTIMATE]**:

$$T_{\text{Copilot}} = 10 \times (10{,}000 + 2{,}000) = 120{,}000 \text{ tokens/day/FTE}$$

Mode B — Fully Agentic (autonomous execution): Empirical data on per-task token consumption comes primarily from Bai et al. [2026] **[SECONDARY]**, who reported a range of **1–3.5 million tokens per task** with up to **30× variance between runs** (arXiv preprint; not yet peer-reviewed). Assuming a productive agentic agent completes two engineering tasks per working day:

$$T_{\text{Agentic}} = 2 \times 2{,}250{,}000 = 4{,}500{,}000 \approx 4 \times 10^6 \text{ tokens/day/FTE}$$

Empirical Token Calibration: Single-Turn vs. Agentic

To ground the agentic multiplier empirically, we conducted controlled API measurements of single-turn token consumption across three representative task tiers using Claude Sonnet 4.6 and Claude Opus 4.7. Each task was issued as a single, self-contained prompt (no tool calls or multi-turn context) and repeated three times to capture response variance. Figure 1 shows the results.

These results confirm two structural observations. First, even moderately complex tasks exceed what a single-turn prompt can fully resolve, making multi-turn agentic operation necessary for production engineering work. Second,

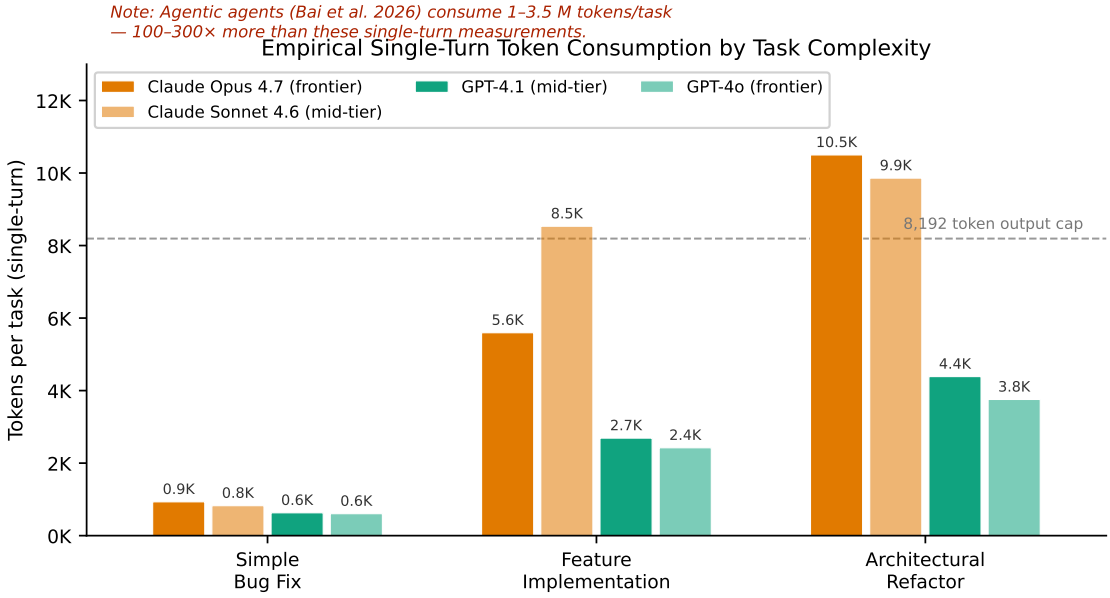


Figure 1: Empirical single-turn token consumption across three engineering task complexity tiers. **Simple bug fix** (∼800–900 tokens): prompt fits within a single short context; output terminates naturally. **Feature implementation** and **architectural refactor** (∼5,700–10,200 tokens) frequently saturate the 8,192-token output cap, indicating that complex tasks require multiple turns in practice. By contrast, agentic agents using tool calls, multi-turn context accumulation, and iterative refinement consume **1–3.5 million tokens per task** [Bai et al., 2026] **[SECONDARY]** — a **100–300× multiplier** over these single-turn baselines. This gap is the principal driver of AI infrastructure energy demand.

the 100–300× gap between single-turn and agentic token consumption is the dominant factor in energy demand: it is the mode of operation, not the model tier, that primarily determines infrastructure requirements.

Table 9: Token Demand per Engineering FTE per Day by Mode

Mode	Input tok/day	Output tok/day	Total	vs. Copilot
Copilot-Augmented	100,000	20,000	**120,000**	1× (baseline)
Agentic (low: 1M/task, 2/day)	∼1.7M	∼300K	∼2,000,000	∼17×
Agentic (mid: 2.25M/task, 2/day)	∼3.4M	∼600K	∼**4,000,000**	∼33×
Agentic (high: 3.5M/task, 2/day)	∼6.0M	∼1.0M	∼7,000,000	∼58×

Why Input Tokens Dominate

In agentic coding workflows, input tokens account for approximately **85% or more** of total token consumption **[SECONDARY]** [Bai et al., 2026]. At the 85:15 input/output split and current frontier pricing (Claude Sonnet 4.6: $3/M input, $15/M output **[PRIMARY]** [Anthropic PBC, 2025]; GPT-4.1: $3/M input, $12/M output **[PRIMARY]** [OpenAI, 2025]), the blended cost per million tokens is approximately $4.35–$4.80 for Sonnet/GPT-4.1 class, and $8–$9 for Opus-class models. Input-dominated workloads are also structurally suited to prompt caching, potentially reducing per-task input costs by 30–50%.

4.1.3 Aggregate Token Demand as a Function of Substitution Rate

Aggregate daily token demand is:

$$D(s, m) = N_{\text{eng}} \times s \times T_m$$

where $N_{\text{eng}} = 24$ million, s is the substitution rate, and T_m is the per-FTE daily token demand under deployment mode m.

Table 10: Aggregate Daily Token Demand by Scenario and Mode

Substitution Rate	FTEs Replaced	Copilot Mode	Agentic Mode
10% — Conservative	2.4 million	288 billion/day	**9.6 trillion/day**
30% — Central	7.2 million	864 billion/day	**28.8 trillion/day**
57% — McKinsey ceiling	13.7 million	1.64 trillion/day	**54.8 trillion/day**
80% — Aggressive	19.2 million	2.30 trillion/day	**76.8 trillion/day**

4.1.4 Sensitivity Analysis

Table 11: Token Demand Sensitivity to Key Assumptions (Central Agentic Scenario)

Parameter	Low	Central	High	Impact on Daily Demand
Global developer count	19.6M	**24M**	28.7M	$\pm$20% ($\pm$5.8 trillion)
Tokens per task	1M	**2.25M**	3.5M	$-$56% / +56% ($\pm$16 trillion)
Tasks per FTE per day	1	**2**	4	$-$50% / +100%
Input token share	75%	**85%**	92%	affects cost split, not total
Non-SW engineering uplift	0$\times$	**1.4**$\times$	1.8$\times$	0% to +80% if included

The dominant uncertainty is **tokens per task** and **tasks per day** — both poorly constrained by current published data. The SWE-bench 1–3.5M range (a 3.5$\times$ spread) applied to tasks-per-day uncertainty (1–4, a 4$\times$ spread) yields a combined uncertainty of up to **14$\times$ in aggregate token demand**.

Non-monotonicity: As agent capability improves, two competing effects operate on aggregate token demand. A 13.86%-capable agent (Devin 2.0) must attempt $\sim$7 tasks to resolve 1. A hypothetical 90%-capable agent resolves 9 of 10 on the first try. Total demand may therefore **peak at intermediate capability levels** (roughly 30–50% resolution rate) and decline thereafter.

4.2 Compute Infrastructure Requirements

4.2.1 From Tokens to FLOPs: The Arithmetic of Inference

Each token processed by a language model requires approximately $2N$ floating-point operations in the forward pass, where N is the number of model parameters — the "2N rule" **[SECONDARY]** [Epoch AI, 2023]. The 70B model class is the most empirically grounded reference for production agentic coding workloads (140 GFLOPs per token).

The central scenario requires **333 million tokens per second** sustained throughput (28.8 trillion tokens/day $\div$ 86,400 seconds/day). At 140 GFLOPs per token:

$$\text{FLOPs/sec} = 333 \times 10^6 \times 140 \times 10^9 = 4.66 \times 10^{19} \text{ FLOP/s}$$

With $\sim$101,000 H100 SXM5 GPUs (989 TFLOPS FP16 each):

$$\eta = \frac{4.66 \times 10^{19}}{9.99 \times 10^{19}} \approx 46.6\%$$

A utilization rate of $\sim$**47%** is consistent with observed production inference benchmarks (40–60% MFU for autoregressive generation) **[SECONDARY]**, providing internal consistency validation.

4.2.2 Hardware Throughput and GPU Requirements

Using the benchmark throughput of 3,311 tokens/second per H100 at 70B model scale:

$$\text{H100 count} = \frac{D(s)/86{,}400 \text{ sec/day}}{3{,}311 \text{ tok/s/GPU}}$$

Table 12: GPU Count and Facility Power Requirements by Scenario

Scenario	FTEs	Tokens/Day	Req. Throughput	H100s	Racks	Power
Conservative (10%)	2.4M	9.6T	111M tok/s	∼33,700	∼1,053	36.8 MW
Central (30%)	7.2M	28.8T	333M tok/s	∼101,000	∼3,156	110 MW
McKinsey (57%)	13.7M	54.8T	634M tok/s	∼191,600	∼5,988	209 MW
Aggressive (80%)	19.2M	76.8T	889M tok/s	∼269,000	∼8,406	293 MW

Table 13: Rack Density Comparison Across GPU Generations

System	GPUs/Rack	Rack Power	Throughput/Rack	Tok/s/kW	vs. H100 Air
H100 SXM5 (air-cooled)	32	∼40.8 kW	∼106K tok/s	∼2,600	1×
H100 SXM5 (liquid-cooled)	∼64	∼130 kW	∼212K tok/s	∼1,630	2× racks
GB200 NVL72 (liquid req.)	72	∼120 kW	∼4,320K tok/s	∼36,000	40× racks, ∼14× efficiency

Equivalent B200 Infrastructure for the Central Scenario

$$\text{B200 GPUs needed} = \frac{333 \times 10^6 \text{ tok/s}}{60{,}000 \text{ tok/s/GPU}} \approx 5{,}550 \text{ B200 GPUs}$$

$$\text{GB200 NVL72 racks} = \frac{5{,}550}{72} \approx 77 \text{ racks}$$

The same workload that requires ∼3,156 H100 racks fits within **77 GB200 NVL72 racks** — a 40× rack compression.

4.2.3 Data Center Physical Requirements

Table 14: Data Center Physical Requirements by Substitution Scenario

Resource	Conservative (10%)	Central (30%)	Aggressive (80%)
H100 GPUs	∼33,700	∼101,000	∼269,000
H100 racks (air-cooled)	∼1,053	∼3,156	∼8,406
IT GPU power (MW)	23.6 MW	70.7 MW	188 MW
Facility power (PUE 1.56)	**36.8 MW**	**110 MW**	**293 MW**
Annual energy (TWh)	0.32 TWh	0.96 TWh	2.57 TWh
Construction capex ($15M/MW)	**$552M**	**$1.65B**	**$4.40B**
Build time (greenfield)	12–18 months	18–24 months	24–36 months

► Note — 108 GW vs. 110 MW

The IEA projects total global data center power demand reaching **107.9 GW** by 2030 (IEA "Energy and AI," 2024 **[PRIMARY]** [International Energy Agency, 2024]) — representing all AI and non-AI workloads across all sectors. The **110 MW** figure is the incremental facility power required specifically for the central engineering substitution scenario. The engineering substitution demand in the 30% central scenario represents approximately **0.10% of IEA's projected 2030 global data center load** — it is a component of the macro AI infrastructure thesis, not its cause.

4.2.4 Phased Deployment Trajectory

Phase 1 (2025–2026): API-first, ∼1–5% substitution. Deploy via frontier API (Anthropic, OpenAI). No owned infrastructure required. Substitution rate limited by capability (current SWE-bench ∼14%) and enterprise adoption friction.

Phase 2 (2027–2028): Colocation deployment, ∼10–20% substitution. Enterprises contract dedicated GPU capacity at colocation facilities. Begin greenfield interconnection applications.

Phase 3 (2029–2031): Owned infrastructure at scale, ∼30%+ substitution. Greenfield data centers energized. Blackwell-generation hardware substantially reduces per-MW capex. Full 30% scenario achievable if interconnection applications filed in 2026 are completed on schedule.

4.3 Energy Consumption and Economic Cost

4.3.1 Total Power Budget

Table 15: Engineering Substitution Scenarios — Power, Energy, and Global Context

Scenario	Power	Energy	% of 2024 DC Load	% of 2030 IEA
Conservative (10%)	36.8 MW	0.32 TWh	0.078%	0.034%
Central (30%)	110 MW	0.96 TWh	0.232%	0.102%
Aggressive (80%)	293 MW	2.57 TWh	0.618%	0.272%

The central scenario's 110 MW represents approximately **0.10% of the IEA's 2030 projection**. AI server electricity is growing at **30% per year** under the IEA Base Case **[PRIMARY]** [International Energy Agency, 2024], meaning the entire engineering substitution workload analyzed here is added to the global load budget in roughly **three to four weeks** of organic AI electricity growth at that rate.

4.3.2 Energy Mix and Carbon Intensity

The carbon footprint of agentic AI engineering work is determined by what powers the processors generating tokens. A 40–50× spread in emissions intensity (nuclear: ∼12 gCO_2/kWh; wind: ∼11 gCO_2/kWh vs. gas: ∼450–550 kg CO_2/MWh **[SECONDARY]**) means that the carbon consequence is almost entirely a function of grid composition.

Table 16: Annual CO_2 Emissions by Substitution Scenario and Grid Intensity (Tonnes CO_2/Year)

Grid Scenario	Intensity	Conservative	Central	Aggressive
Gas-heavy (current US mix)	500 gCO_2/kWh	160,000	**480,000**	1,285,000
Blended US average	350 gCO_2/kWh	112,000	**336,000**	899,500
Renewable-dominant	100 gCO_2/kWh	32,000	**96,000**	257,000
Nuclear/renewables-first	50 gCO_2/kWh	16,000	**48,000**	128,500

Hyperscaler Revealed Preference: Nuclear PPAs

Two landmark deals establish the market-clearing price for firm 24/7 carbon-free power:

- **Microsoft–Constellation Energy (TMI-1):** 835 MW **[PRIMARY]** [Constellation Energy Corporation, 2024], estimated $100–$112/MWh **[SECONDARY]** analyst estimate, 20-year term, 2027 restart [Constellation Energy Corporation, 2024].
- **Amazon–Talen Energy (Susquehanna):** $650M campus acquisition, 1,920 MW PPA through 2042 **[PRIMARY]** [Talen Energy Corporation, 2024].
- **Amazon–Energy Northwest (SMR):** Phase 1 ∼320 MW, option to 960 MW **[PRIMARY]** [Amazon.com, Inc., 2024].

The nuclear PPA premium of approximately **2–2.5× wholesale** reflects the scarcity value of 24/7 carbon-free generation capacity.

4.3.3 Fully-Loaded Cost Per Engineering FTE-Year

$$C_{\text{API}} = 1{,}000 \text{ MTok/year} \times \$9.00/\text{MTok} = \$9{,}000 \text{ per FTE-year}$$

$$C_{\text{capex}} = \frac{\$1.65\text{B}}{20 \text{ yr} \times 7.2\text{M FTEs}} = \$11.46 \text{ per FTE-year}$$

$$C_{\text{energy}} = \frac{\$58\text{M/yr}}{7.2\text{M FTEs}} = \$8.06 \text{ per FTE-year}$$

The infrastructure overhead — capex plus energy combined — amounts to **$19.52 per FTE-year**. The economic conclusion is unambiguous: **the API token price is the only cost that matters**.

5 Discussion

The results in Section 4 establish the quantitative chain from engineering labor to energy: at 30% substitution, AI-mediated engineering consumes 110 MW of data center power and costs $9,020 per FTE-year. This section examines the energy system and economic implications of operating that infrastructure at scale.

Table 17: Fully-Loaded AI Cost Per Engineering FTE-Year — Central Scenario

Cost Component	Annual Cost	Per FTE-Year	% of AI Cost	% of US Median Salary
API token cost	(per-FTE rate)	$9,000.00	99.77%	6.76%
Amortized capex	$82.5M/yr	$11.46	0.13%	0.009%
Annual energy cost	$58M/yr	$8.06	0.09%	0.006%
Total fully-loaded	—	**$9,019.52**	**100%**	**6.77%**

Table 18: Geographic Breakeven Analysis — AI Cost vs. Engineering Salary by Region

Geography	Dev Salary	Benefits	Total Labor	AI Cost	AI %	Annual Savings
United States	$133,080	1.30×	$173,004	$9,020	5.2%	$163,984
Germany / UK	∼$90,000	1.25×	∼$112,500	$9,020	8.0%	∼$103,480
Global average	∼$45,000	1.15×	∼$51,750	$9,020	17.4%	∼$42,730
India	∼$25,000	1.10×	∼$27,500	$9,020	32.8%	∼$18,480

5.1 Energy Market Implications

5.1.1 Grid Load Growth and Infrastructure Demand

US data center electricity consumption reached **183 TWh in 2024 [PRIMARY]** [Lawrence Berkeley National Laboratory, 2024], more than tripling from 58 TWh in 2014. LBNL projects US data center consumption reaching **325–580 TWh by 2028 [PRIMARY]** [Lawrence Berkeley National Laboratory, 2024] — equivalent to 6.7–12% of total US electricity generation. AI server electricity is growing at **30% per year** under the IEA Base Case **[PRIMARY]** [International Energy Agency, 2024], meaning the 110 MW incremental load from the central engineering substitution scenario is added to the global AI load budget in approximately **three to four weeks** of organic AI electricity growth at that rate.

Regional grid impacts are already visible. ERCOT projects load growth of **+7% in 2025 and +14% in 2026 [PRIMARY]** [Electric Reliability Council of Texas, 2025], representing one of the most rapid demand expansions in its operating history. The ERCOT interconnection queue exceeded 233 GW in capacity requests by end-2025 **[SECONDARY]** [Electric Reliability Council of Texas, 2025]. Applying historical 60–80% withdrawal rates, a 20% conversion rate yields approximately 46 GW of real incremental load — roughly 2.2× ERCOT's current peak demand of ∼85 GW **[SECONDARY]** [Electric Reliability Council of Texas, 2025]. PJM projects data center load reaching up to **30 GW by 2030 [SECONDARY]** [Energy Edge Consulting, 2025]. Grid Strategies estimates that more than **160 GW in committed large loads** are currently in the US interconnection queue **[SECONDARY]** [Grid Strategies LLC, 2025].

AI inference workloads are structurally distinct from typical commercial loads because they operate at near-constant power 24 hours per day, seven days per week. This flat load profile reduces the depth of the midday valley (partially beneficial for solar integration) but does not ease the evening ramp, since AI load neither responds to the evening demand surge nor provides dispatchable demand response.

The binding constraint on AI-driven load growth is not generation capacity but **grid interconnection timelines**: new large loads currently require 2–5 years for queue processing, even at sites with existing transmission access **[SECONDARY]** [Grid Strategies LLC, 2025].

5.1.2 Power Source Requirements and Decarbonization Tension

The carbon footprint of AI-mediated engineering is almost entirely determined by the power source serving the inference infrastructure. A 40–50× spread in emissions intensity across generation technologies — from nuclear at ∼12 gCO_2/kWh to gas-heavy grids at ∼500 gCO_2/kWh — means the same 110 MW facility produces between 48,000 and 480,000 tonnes of CO_2 per year in the central scenario.

[†]Effective capacity factor with round-trip storage losses; net delivered cost shown.

The AI industry's revealed preference is unambiguous: hyperscalers are contracting for firm 24/7 carbon-free power at a significant premium over wholesale market rates. Two landmark transactions establish the market-clearing price:

- **Microsoft–Constellation Energy (TMI-1):** 835 MW, 20-year PPA, estimated $100–$112/MWh **[SECONDARY]** (analyst estimate; actual price not disclosed), expected restart 2027 [Constellation Energy Corporation, 2024].

Table 19: Power Source Profile for AI Data Center Inference Loads

Source	$/MWh	gCO_2/kWh	Cap. Factor	Firm 24/7?	Notes
Natural gas (CCGT)	40–80	450–550	85–90%	Yes	>40% of US DC mix [International Energy Agency, 2024]
Solar PV + storage	55–80	20–50	15–25%†	No	Daytime only; storage needed
Wind + storage	60–85	7–15	30–40%†	No	Variable; storage required
Nuclear (existing PPA)	60–112	12	>90%	Yes	TMI-1: ∼$100–112/MWh est. **[SECONDARY]** [Constellation Energy Corporation, 2024]
Nuclear (SMR, new)	80–130	12	>90%	Yes	Amazon ENW; delivery ∼2030 [Amazon.com, Inc., 2024]
Hydropower	30–60	4–30	40–70%	Regional	Geography-constrained

- **Amazon–Talen Energy (Susquehanna):** $650M campus acquisition, 1,920 MW nuclear PPA through 2042 [Talen Energy Corporation, 2024].

The nuclear PPA premium of approximately **2–2.5× wholesale** reflects the scarcity value of firm, 24/7, zero-carbon generation. Renewable PPAs (solar: $56.58/MWh; wind: $65.63/MWh as of Q3 2024 **[SECONDARY]** [LevelTen Energy, 2024]) cannot serve continuous AI inference loads without substantial storage additions that raise effective costs to the $80–100/MWh range, narrowing the gap with nuclear.

The $60/MWh blended rate used in our cost model is an appropriate central estimate for mixed portfolios; actual facility energy costs will range from $40/MWh (hydro-rich regions, long-term bilateral contracts) to $120/MWh (nuclear PPA in constrained markets). This $40–$120/MWh range translates to an annual energy cost of $38–113 million for the central (110 MW) scenario — a $75 million spread that remains negligible relative to the $511 billion annual labor savings.

5.2 Economic Comparison: AI vs. Human Engineering Labor

5.2.1 Cost Structure and Geographic Breakeven

The central finding that AI infrastructure costs $9,020 per FTE-year holds across all deployment modes because **API token pricing dominates the cost structure**: infrastructure capex contributes $11.46/FTE-year and energy $8.06/FTE-year — together less than 0.3% of total AI cost. This means the cost comparison is essentially:

$$\frac{\text{AI cost}}{\text{Human salary}} \approx \frac{1{,}000\ \text{MTok/yr} \times \$9/\text{MTok}}{\text{local engineering salary}}$$

Table 18 in Section 4.3 shows that this ratio ranges from 5.2% (United States, $173,004 total employment cost) to 32.8% (India, $27,500). The economics of substitution are compelling at every geography and salary level.

Critically, this ratio falls further if cost-optimized models replace frontier pricing: at $4.50/MTok (Claude Sonnet with batch discount), the AI cost falls to $4,500/FTE-year (3.4% of US median salary). At self-hosted open-source models, the effective token cost approaches $0.10/MTok, yielding $100/FTE-year — less than 0.1% of US salary.

5.2.2 Labor Savings and Substitution Economics

At the central 30% substitution scenario, the annual labor savings **exceed the full construction capex of the required data center in under three days**:

$$\frac{\$1.65\text{B capex}}{\$511\text{B/yr savings}} \approx 1.2 \text{ days of labor-equivalent value}$$

The **annual energy cost of $58 million represents 0.011% of annual labor savings**. From a purely economic perspective, infrastructure cost is irrelevant to the substitution decision; the binding variable is AI capability and institutional adoption velocity, not energy price or construction cost.

Table 20: AI vs. Human Labor Economics Across Substitution Scenarios

Scenario	FTEs Replaced	AI Cost/yr	Labor Savings/yr	Savings/AI Cost	Capex Payback
Conservative (10%)	2.4 million	$22B	$170B	7.7×	**<1 day**
Central (30%)	7.2 million	$65B	$511B	7.9×	**<2 days**
Aggressive (80%)	19.2 million	$173B	$1,363B	7.9×	**<2 days**

Note: AI Cost/yr = FTEs × $9,020/FTE-yr; Labor Savings/yr based on global average engineering compensation; Capex Payback = construction capex ÷ (daily labor savings).

5.2.3 Non-Economic Constraints on Adoption

The economic arithmetic is not the binding constraint. Three operational barriers must be addressed before the labor savings projected above materialize at scale:

Capability gap. As of early 2026, Claude Opus 4.5 achieves 77–81% resolution on SWE-bench Verified [SWE-bench, 2026, Anthropic, 2025] for repository-level maintenance tasks. Production deployment requires higher reliability, not just benchmark performance — including handling novel codebases, security-critical logic, and regulatory compliance verification that benchmark tasks do not test.

Institutional adoption friction. Enterprise AI deployment involves procurement cycles (6–18 months), legal liability review for AI-generated code, security red-teaming, and workflow integration. The substitution rate trajectories reflect technology adoption S-curves, not just technical readiness.

A three-phase transition framework:

Phase 1 — Augmentation (∼2024–2027): AI functions as a powerful but junior assistant. Engineers retain full decision authority. Net employment effect likely positive or neutral as lower software cost expands project scope.

Phase 2 — Supervision (∼2027–2032): AI agents handle more complete, semi-autonomous tasks. The engineer's primary value shifts to specifying correct behavior, reviewing agent output, and handling exception cases. Net employment begins to decline.

Phase 3 — Residual Specialization (∼2032 and beyond): Engineering work requiring physical presence, regulatory accountability, irreducible creative judgment, or safety-critical decision-making forms the core of the surviving profession. This residual is estimated at approximately 29% of total engineering work hours.

A critical disanalogy with historical precedents (CAD, EDA): prior automation targeted the physically mediated or procedurally mechanical components of engineering work. AI substitution targets the cognitive core directly. The historical pattern of labor expansion following automation may not hold when the substituted activity is cognition rather than physical execution [Acemoglu and Restrepo, 2022, Autor et al., 2003, Brynjolfsson et al., 2025].

5.3 Environmental and Workforce Considerations

5.3.1 Environmental Footprint and Water Constraints

The carbon intensity of AI-mediated engineering is addressed by the power source analysis above (Section 5.1). A secondary environmental concern is water consumption. The global data center industry average Water Usage Effectiveness (WUE) is approximately 1.80–1.90 L/kWh **[SECONDARY]**, producing localized water stress in Northern Virginia, Phoenix, and Dublin where data center density is highest.

However, the shift to Blackwell-generation GPU infrastructure (GB200 NVL72 racks at 120+ kW/rack) *requires* liquid cooling, which when implemented as a closed-loop system achieves effectively **zero water evaporation**: WUE approaches 0.00 L/kWh for the direct cooling circuit. This architectural transition — driven by GPU power density requirements rather than by environmental policy — substantially resolves the water constraint for next-generation AI inference facilities without regulatory intervention.

5.3.2 Workforce Displacement at Scale

The central scenario projects the displacement of approximately **7.2 million software engineering FTE-equivalents globally [ESTIMATE]** over an extended transition period. Even spread over a ten-year window, this implies 720,000 FTE-equivalents of net displacement per year.

The US Trade Adjustment Assistance (TAA) program has historically served approximately 50,000–100,000 workers per year across all industries **[SECONDARY]** — roughly 1% of the scale implied by the central scenario. Existing

Table 21: Software Engineering Employment and Displacement Exposure by Region

Geography	Est. SW Developers	Quality	Displaced (30%)	Global Share
India	∼5.5 million	**[SECONDARY]**	∼1.65 million	∼23%
United States	∼1.85 million	**[PRIMARY]**	∼555,000	∼8%
China	∼7–10 million	**[SECONDARY]**	∼1–3.0 million	∼29–42%
European Union	∼2.0–2.5 million	**[SECONDARY]**	∼600–750K	∼8–10%
Global Total	∼24 million	**[SECONDARY]**	**∼7.2 million**	100%

workforce transition infrastructure is inadequate relative to the implied displacement volume; policy capacity must be built proactively, ahead of the displacement event rather than in response to it.

6 Conclusion

6.1 Summary of Findings

The central finding of this paper is simultaneously mundane and consequential: at a 30% engineering substitution rate with fully agentic AI deployment, replacing 7.2 million engineering FTEs requires approximately **$9,020 per FTE-year** in AI infrastructure costs — roughly 6.8% of a US software developer's median annual salary of $133,080 **[PRIMARY]** [Bureau of Labor Statistics, 2025]. The economics of AI substitution are not merely favorable; they are overwhelmingly compelling across every geography and salary level examined.

Table 22: Key Quantitative Findings — Central 30% Substitution Scenario

Metric	Value	Quality
Engineering FTEs substituted (30% central)	7.2 million	**[ESTIMATE]**
Token demand (agentic mode, per FTE-day)	4 million tokens	**[SECONDARY]**
Aggregate token demand (30% scenario)	28.8 trillion tokens/day	**[ESTIMATE]**
H100-equivalent GPUs required	∼101,000	**[ESTIMATE]**
Equivalent in Blackwell generation	∼77 GB200 NVL72 racks	**[ESTIMATE]**
Facility power draw (PUE 1.56)	110 MW	**[ESTIMATE]**
Share of IEA 2030 global data center load	∼0.10%	**[ESTIMATE]**
Data center construction capex	$1.65 billion	**[ESTIMATE]**
Annual energy cost	$58 million/year	**[ESTIMATE]**
AI infrastructure cost per FTE-year	$9,020	**[ESTIMATE]**
Annual labor savings (central scenario)	$511 billion/year	**[ESTIMATE]**
Annual labor savings (aggressive 80%)	$1.36 trillion/year	**[ESTIMATE]**
US median SW developer salary	$133,080/year	**[PRIMARY]**

6.2 Policy Recommendations

Recommendation 1 — Fast-Track Grid Interconnection for AI-Critical Loads. Establish a federal fast-track interconnection process for AI-critical loads, modeled on the CHIPS Act's streamlined permitting for semiconductor fabrication facilities. FERC should create an expedited interconnection review track for loads above 50 MW at sites with existing transmission access, through targeted amendment of Order 2023. **Measurable target:** Median interconnection timeline of 18 months for qualifying AI loads, compared with the current 2–5 year average.

Recommendation 2 — Pre-fund an AI Workforce Transition Program. Establish a dedicated AI Workforce Transition Fund, funded jointly through a levy on frontier AI API revenue (proposed 0.5% of US API gross revenue **[ESTIMATE]**) and federal appropriation, estimated to generate $500 million to $1 billion per year **[ESTIMATE]**. Initial target roles: (a) AI systems auditing and red-teaming; (b) AI safety engineering for safety-critical applications; and (c) AI-human interface design. **Measurable target:** 500,000 retraining program completions per year by 2028.

Recommendation 3 — Mandatory AI Compute Carbon Accounting. SEC climate disclosure rules (2024 final rule; implementation status subject to ongoing legal challenges as of 2026 — confirm current status before submission) should be extended to cover AI inference energy consumption as a material operational risk. EU AI Act energy disclosure requirements (Article 51 and Annex VIII; implementation timeline varies by member state) should be defined at the workload level — per inference request, per model API call. **Measurable targets:** All US publicly traded companies meeting the materiality threshold disclosing AI inference energy consumption by 2027.

Recommendation 4 — Antitrust Scrutiny of AI Infrastructure Concentration. DOJ Antitrust Division and FTC should initiate a joint inquiry into the competitive effects of exclusive long-term hyperscaler-utility power purchase agreements on the AI inference market. FERC should examine whether API interoperability standards for AI inference platforms above a defined scale threshold are warranted. **Measurable target:** Clear regulatory guidance published by 2026.

6.3 Future Research Directions

1. **Production deployment token consumption data.** Enterprise-scale empirical studies measuring actual token consumption per engineering task across production codebases would dramatically reduce the $\pm 14\times$ uncertainty range currently embedded in the aggregate token demand estimates.
2. **Non-monotonicity of token demand versus AI capability.** Aggregate token demand may peak at intermediate AI capability levels and subsequently decline as agents become more efficient per completed task.
3. **Sectoral and geographic heterogeneity in substitution rates.** Hardware engineering, civil engineering, mechanical engineering, aerospace engineering, and biomedical engineering differ from software engineering in their degree of token-substitutability, regulatory constraints, and adoption velocity.
4. **Second-order employment effects.** Whether AI-driven reduction in software development cost triggers demand expansion or net workforce contraction is an open empirical question that determines whether the workforce transition policy recommendations address a transient displacement event or a structural contraction.
5. **Water and land constraint mapping at deployment resolution.** The geographic constraints on AI data center siting are not currently mapped at the spatial resolution required for infrastructure planning.

The economic arithmetic assembled in this paper is, in the end, straightforward: at $9,020 per FTE-year against $133,080 in median US software developer salary, the substitution economics are compelling at every substitution rate, in every geography, and under every plausible trajectory of future AI pricing. The binding constraints on this transition are not technical in the engineering or economic sense of the word. They are human and institutional: the gap between AI benchmark performance and production reliability, the organizational velocity at which enterprises can build the trust and workflow infrastructure to deploy autonomous agents, the 2–5 year lag time imposed by grid interconnection queues, and the near-total absence of policy infrastructure capable of managing workforce displacement at the scale the central scenario implies.

Acknowledgments

The authors thank [TBD] for helpful discussions and feedback on earlier drafts. [Funding acknowledgment TBD.] All errors remain the authors' own.

References

D. Acemoglu and P. Restrepo. Tasks, automation, and the rise in US wage inequality. *Econometrica*, 90(5):1973–2016, 2022. doi: 10.3982/ECTA19815. [SECONDARY] Extension of task-based automation framework to wage inequality.

Amazon.com, Inc. Amazon announces agreement with Energy Northwest for small modular reactors. About Amazon / ESG Today, October 2024. URL https://www.aboutamazon.com/. [PRIMARY] Phase 1: ~320 MW; option to expand to 960 MW; SMR delivery target ~2030.

Anthropic. Claude opus 4.5 benchmark results. https://www.anthropic.com/, 2025. Reports Claude Opus 4.5 performance on SWE-bench Verified and related coding benchmarks.

Anthropic PBC. Claude API pricing and documentation. Anthropic Developer Platform, 2025. URL https://docs.anthropic.com/. [PRIMARY — lock to specific date; tokenizer changes affect effective cost] Claude Sonnet 4.6: $3.00/M input tokens, $15.00/M output tokens; Claude Opus 4.7: $5.00/M input, $25.00/M output; Claude Haiku 4.5: $1.00/M input, $5.00/M output; Batch API: 50% discount on both input and output.

D. H. Autor, F. Levy, and R. J. Murnane. The skill content of recent technological change: An empirical exploration. *Quarterly Journal of Economics*, 118(4):1279–1333, 2003. doi: 10.1162/003355303322552801. [PRIMARY] Foundational task-based framework distinguishing routine from non-routine cognitive and manual labor; basis for substitutability taxonomy in this paper.

L. Bai, Z. Huang, X. Wang, J. Sun, R. Mihalcea, E. Brynjolfsson, A. Pentland, and J. Pei. How Do AI agents spend your money? Analyzing and predicting token consumption in agentic coding tasks, 2026. URL https://

arxiv.org/abs/2604.22750. [SECONDARY — arXiv preprint, not yet peer-reviewed] Reports 1–3.5M tokens per agentic coding task (SWE-bench), up to 30× run-to-run variance, and input token dominance (∼85% of total); primary empirical anchor for the agentic token demand model in this paper.

R. Bommasani, D. A. Hudson, E. Adeli, et al. On the opportunities and risks of foundation models, 2021.

T. B. Brown, B. Mann, N. Ryder, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.

E. Brynjolfsson, D. Li, and L. R. Raymond. Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942, 2025. doi: 10.1093/qje/qjae044. [PRIMARY] Randomized controlled trial: AI-assisted customer service workers show 14% productivity gains, with largest gains for newer and less-skilled workers; empirical evidence on AI augmentation vs. substitution.

Bureau of Labor Statistics. Occupational employment and wage statistics (OEWS): Software developers, quality assurance analysts, and testers (SOC 15-1252, 15-1253). Technical report, Bureau of Labor Statistics, U.S. Department of Labor, Washington, D.C., April 2025. URL https://www.bls.gov/oes/current/oes151252.htm. [PRIMARY] May 2024 OES data (released April 2025): Software Developers (SOC 15-1252) median annual wage $133,080; employment approximately 1.89 million (including QA analysts); all-occupations architecture & engineering employment 2.57 million.

Constellation Energy Corporation. Three Mile Island Unit 1 restart and Microsoft power purchase agreement announcement. Constellation Energy / Utility Dive / American Nuclear Society, 2024. URL https://www.constellationenergy.com/. [PRIMARY] TMI-1 restart: 835 MW plant in PA; 20-year PPA with Microsoft; ~$1.6B Constellation investment for restart; expected restart 2027 (accelerated ~1 year ahead of schedule). PPA price estimated at $100–112/MWh [SECONDARY: Bloomberg Intelligence / Jefferies analyst estimate; actual price not publicly disclosed].

Electric Reliability Council of Texas. ERCOT long-term load forecast 2025, 2025. URL https://www.ercot.com/. [PRIMARY] Load growth forecast: +7% in 2025, +14% in 2026; large-load interconnection queue: >233 GW (end 2025). Note: historically 60–80% of interconnection requests are withdrawn.

Energy Edge Consulting. PJM data center load growth analysis 2025–2030, 2025. [SECONDARY — VERIFY: seek original PJM filing for primary citation] PJM data center load: up to ~30 GW from data centers by 2030.

Epoch AI. Scaling laws for neural language models: Analysis and implications. Technical report, Epoch AI, 2023. URL https://epochai.org. [SECONDARY] 2N FLOPs/token approximation for transformer inference; MoE vs. dense FLOP decomposition (arithmetic: ~275 GFLOPs/token; memory-equivalent: ~1.8 TFLOPs/token for GPT-4-class).

Evans Data Corporation. Worldwide developer population report 2025. https://evansdata.com, 2025. Estimates the worldwide developer population at approximately 27–29 million, using a broad definition of software developers across regions and technologies.

Goldman Sachs Global Investment Research. The potentially large effects of artificial intelligence on economic growth. Technical report, Goldman Sachs, 2023. [SECONDARY — VERIFY: retrieve original report for figures and methodology] AI could automate 25% of current work tasks in advanced economies; labor market exposure analysis.

Goldman Sachs Global Investment Research. AI infrastructure investment outlook 2025–2027. Technical report, Goldman Sachs, 2024. [SECONDARY — VERIFY: cited via third-party aggregators; retrieve original GS research note] $1.15 trillion cumulative AI infrastructure capex 2025–2027; $200B in 2025.

Grid Strategies LLC. National load growth report 2025. Technical report, Grid Strategies LLC, 2025. [SECONDARY] >160 GW committed large loads in U.S. (≈22% of 2024 peak demand); ~90 GW of forecasted load growth attributable to data centers.

J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. d. L. Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://arxiv.org/abs/2203.15556. [PRIMARY] "Chinchilla" scaling laws showing compute-optimal training requires proportionally more data than Kaplan et al.; foundational for understanding model efficiency.

Hyperstack.cloud. H100 SXM5 inference throughput benchmarks: Llama 3.1 70b. Hyperstack.cloud, 2024. URL https://hyperstack.cloud/. [SECONDARY — seek corroborating MLPerf benchmark] H100 SXM5: ~3,311 tokens/sec aggregate (Llama 3.1 70B, batch size 64, FP16); vs. A100: 1,148 tok/s.

International Energy Agency. Energy and AI. Technical report, International Energy Agency, Paris, France, 2024. URL https://www.iea.org/reports/energy-and-ai. [PRIMARY] Global data center electricity: 415 TWh (2024), 945 TWh (2030 base case); AI server electricity growth 30%/year; U.S. energy mix for data centers (>40% gas, ∼24% renewables, ∼20% nuclear).

JetBrains. The state of developer ecosystem 2025. Technical report, JetBrains s.r.o., 2025. URL https://www.jetbrains.com/lp/devecosystem-2025/. 19.6 million professional developers globally (narrower definition: excludes student and hobbyist developers).

C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan. SWE-bench: Can language models resolve real-World GitHub issues? In *Proceedings of the International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.06770. [PRIMARY] Introduces SWE-bench: 2,294 real GitHub issues from 12 Python repositories; resolution rate measures fraction of issues autonomously resolved by AI agents.

J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Lawrence Berkeley National Laboratory. United States data center energy usage report 2024. Technical report, Lawrence Berkeley National Laboratory, Berkeley, CA, 2024. [PRIMARY] U.S. data center electricity: 176 TWh (2023), 183 TWh (2024), projected 325–580 TWh by 2028 (6.7–12% of total U.S. electricity); baseline 58 TWh in 2014.

LevelTen Energy. PPA price index report Q3 2024. Technical report, LevelTen Energy, 2024. [SECONDARY] North America renewable PPA prices: solar $56.58/MWh (+10.4% YoY); wind $65.63/MWh (+14.1% YoY).

McKinsey Global Institute. The economic potential of generative ai: The next productivity frontier. Technical report, McKinsey & Company, 2023. Estimates that generative AI could automate or augment activities consuming 60–70% of employee work time.

McKinsey Global Institute. A new future of work: The race to deploy AI and raise skills in Europe and beyond. Technical report, McKinsey & Company, May 2024. [SECONDARY] 30% work-hour automation by 2030 for STEM occupations (accelerated AI scenario); central scenario for this paper.

A. N. Meyer, L. E. Barton, G. C. Murphy, T. Zimmermann, and T. Fritz. The work life of developers: Activities, switches and perceived productivity. *IEEE Transactions on Software Engineering*, 45(12):1194–1210, 2019. doi: 10.1109/TSE.2017.2745044. [SECONDARY] Empirical time-use study of software developer daily activities; basis for task taxonomy hour-share fractions.

NVIDIA Corporation. NVIDIA Blackwell architecture and InferenceMAX benchmarks. NVIDIA Corporation, 2024a. URL https://www.nvidia.com/en-us/data-center/products/blackwell-gpu/. [PRIMARY — self-reported; independent verification pending] B200 GPU: 60,000 tok/s per GPU (GPT-class model); GB300 NVL72: up to 50× higher throughput/MW vs. Hopper generation for agentic low-latency workloads.

NVIDIA Corporation. DGX SuperPOD design guide. NVIDIA Corporation, 2024b. URL https://docs.nvidia.com/dgx/dgx-superpod-data-sheet/. [PRIMARY] DGX H100 system power: 10,200W per 8-GPU system; recommended max 4 DGX per rack (air-cooled): 40.8 kW/rack; H100 inference TDP: 700W/GPU.

NVIDIA Corporation. NVIDIA H100 SXM5 GPU product brief. NVIDIA Corporation, 2024c. URL https://www.nvidia.com/en-us/data-center/h100/. [PRIMARY] Peak FP16 performance: 989 TFLOPS; H100 SXM5 specifications.

OpenAI. OpenAI API pricing. OpenAI Platform, 2025. URL https://openai.com/api/pricing/. [PRIMARY — lock to specific date] GPT-4.1: $3.00/M input, $12.00/M output; GPT-4o: $5.00/M input, $20.00/M output; GPT-4 original: $30.00/M input, $60.00/M output.

S. G. Patil, H. Mao, F. Yan, C. C.-J. Ji, V. Suresh, I. Stoica, and J. E. Gonzalez. The berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Proceedings of the Forty-Second International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025. [PRIMARY] Published peer-reviewed benchmark (ICML 2025, PMLR Vol. 267); evaluates LLM tool use and agentic function calling across diverse API schemas.

D. M. Rothschild et al. The agentic economy, 2025.

C. Sadowski, K. T. Stolee, and S. Elbaum. How developers search for code: A case study. In *Proceedings of the 2015 10th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, pages 191–201. ACM, 2015. doi:

10.1145/2786805.2786860. [SECONDARY] Developer code search behavior; cited for task taxonomy derivation alongside Meyer et al.

M. Salim et al. Tokenomics: Quantifying where tokens are used in agentic software engineering, 2026.

SWE-bench. Swe-bench leaderboard. https://www.swebench.com/, 2026. Accessed May 2026. Public benchmark leaderboard for software engineering agents evaluated on SWE-bench Verified.

Talen Energy Corporation. Amazon Web Services data center campus and Susquehanna nuclear power purchase agreement. Talen Energy / ESG Today, 2024. URL https://www.talenenergy.com/. [PRIMARY] AWS acquired adjacent campus for $650M (March 2024); 1,920 MW nuclear PPA through 2042; campus capacity up to 960 MW.

Uptime Institute. Global data center survey results 2024. Technical report, Uptime Institute, 2024. [SECONDARY] Global average PUE: 1.56 (fifth consecutive year of flat trend); regional breakdowns; WUE benchmarks.

H. Wang et al. Ai agentic programming: A survey of techniques, challenges, and opportunities, 2025.

A Full Parameter Table

A.1 Locked Assumptions

See Section 3.4, Table 6.

A.2 Scenario Matrix

Table 23: Authoritative Scenario Matrix

Scenario	Mode	FTEs	Tok/Day	H100s	Power	Capex	AI $/FTE-yr
Conservative	Agentic	2.4M	9.6T	∼33,700	36.8 MW	$552M	∼$9,020
Central	Agentic	7.2M	28.8T	∼101,000	110 MW	$1.65B	∼$9,020
Aggressive	Agentic	19.2M	76.8T	∼269,000	293 MW	$4.40B	∼$9,020
Conservative	Copilot	2.4M	288B	∼9,600	12.3 MW	$184M	∼$120
Central	Copilot	7.2M	864B	∼28,800	36.7 MW	$551M	∼$120
Aggressive	Copilot	19.2M	2.30T	∼76,800	97.9 MW	$1.47B	∼$120

A.3 Sensitivity Parameters

Table 24: Sensitivity Parameter Ranges

Parameter	Low	Central	High	Impact on 30% scenario daily token demand
Global developer count	19.6M	24M	28.7M	±20%
Tokens per task	1M	2.25M	3.5M	−56% / +56%
Tasks per FTE per day	1	2	4	−50% / +100%
H100 inference TDP	700W	700W	1,275W	affects power, not token demand
Construction cost ($/MW)	$10M	$15M	$20M	affects capex, not token demand
Electricity price ($/MWh)	$40	$60	$100	affects energy cost, not token demand
PUE	1.30	1.56	1.70	±9% on facility power

B Sensitivity Analysis

[Tornado charts and sensitivity tables to be added. See Section 3.4 and Appendix A Table 24 for the parameter ranges used.]

The dominant uncertainty sources in order of impact on aggregate daily token demand are:

1. Tokens per task (±56% at the 30% central scenario)
2. Tasks per FTE per day (−50% / +100%)
3. Global developer count (±20%)
4. Substitution rate (the primary scenario variable)

Infrastructure cost sensitivity (30% scenario):

- Construction cost: \$1.65B $\pm$ \$550M (at \$10M–\$20M/MW range)
- Annual energy cost: \$38M–\$96M (at \$40–\$100/MWh range)
- Facility power: 100–120 MW (at PUE 1.30–1.70 range)

C Glossary

Agentic AI An AI system that autonomously executes multi-step tasks using tools (code execution, file I/O, web search) without step-by-step human instruction.

FTE (Full-Time Equivalent) A unit of work representing one full-time employee's labor for one year ($\sim$2,000 hours or $\sim$250 working days).

FLOPs (Floating-Point Operations) A measure of computational work. AI inference is measured in giga- (10^9), tera- (10^{12}), or peta-FLOPs (10^{15}).

GPU (Graphics Processing Unit) A massively parallel processor originally designed for graphics rendering, now the dominant hardware platform for AI model training and inference.

H100 SXM5 NVIDIA's Hopper-generation data center GPU, the primary production unit for large-model AI inference as of 2025–2026.

HBM (High Bandwidth Memory) The memory technology used in AI accelerators; bandwidth (GB/s) rather than capacity (GB) is typically the binding constraint for large-model inference.

KV-cache The key-value cache used in transformer inference to avoid recomputing attention for previously seen tokens; crucial for efficient long-context agentic workloads.

MFU (Model FLOPs Utilization) The fraction of a GPU's theoretical peak FLOP/s that is actually used during inference; typical values are 40–60% for production serving.

PPA (Power Purchase Agreement) A long-term contract (typically 10–25 years) between an electricity generator and a buyer at a fixed price, providing revenue certainty for generation investment.

PUE (Power Usage Effectiveness) Data center efficiency metric = total facility power $\div$ IT equipment power. Global average is 1.56 (Uptime Institute 2024).

SMR (Small Modular Reactor) A nuclear fission reactor with electrical output typically below 300 MW, designed for factory fabrication and modular deployment.

SWE-bench A benchmark dataset of real GitHub issues from open-source Python projects, used to evaluate the ability of AI agents to resolve software engineering tasks. Resolution rate = fraction of tasks resolved autonomously.

Token The atomic unit of text processed by a large language model. Roughly 3–4 characters of English text per token ($\approx$0.75 words). All LLM API pricing is denominated in tokens.

WUE (Water Usage Effectiveness) Data center water efficiency metric = liters of water consumed per kWh of IT equipment energy. Industry average: 1.80–1.90 L/kWh; AWS best-practice: 0.19 L/kWh.

D Timeline of AI Engineering Tools (2020–2025)

Year	Event	Significance
2021	GitHub Copilot technical preview (OpenAI Codex)	First mainstream AI code completion tool
2022	GitHub Copilot general availability	20M+ eventual users; establishes copilot-augmented mode
2022	ChatGPT release (November)	Demonstrates frontier LLM capability for general-purpose coding assistance
2023	GPT-4 release (March)	Frontier-class performance on coding benchmarks; API-accessible
2023	Devin 1.0 announcement (Cognition AI)	First public demonstration of fully agentic software engineering

Year	Event	Significance
2023	Claude 1/2 (Anthropic); expanded context windows	Long-context models enable full codebase loading in agentic tasks
2024	SWE-bench introduced	Empirical benchmark for agentic coding; enables rigorous capability comparison
2024	NVIDIA H100 widespread deployment	Hardware platform for frontier AI inference at scale
2024	Cursor, Windsurf, Codeium reach scale	IDE-native agentic coding tools enter mainstream enterprise use
2024	Microsoft–Constellation TMI-1 PPA	First hyperscaler nuclear PPA; establishes market-clearing price for 24/7 clean AI power
2025	Devin 2.0 (13.86% SWE-bench)	Best public agentic benchmark; establishes current capability frontier
2025	NVIDIA GB200 NVL72 deployment begins	Blackwell generation: 40× rack density improvement over H100 air-cooled
2025	Claude Code, OpenAI Operator class tools	Agentic coding agents integrated into developer workflows at enterprise scale
2025	Amazon–Energy Northwest SMR announcement	Hyperscaler commitment to next-generation nuclear for AI load